

**EX.NO.:** 09

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## SELF ATTENTION MECHANISM

### AIM

Implement a simple self-attention mechanism to compute attention scores for words in a sentence.

### DATASET DESCRIPTION

The dataset consists of a single sentence: "The quick brown fox jumped over the lazy dog."

The sentence is tokenized into individual words, and each word is assigned a random embedding.

### TECHNIQUES USED

- Tokenization using NLTK
- Random embedding initialization
- Query, Key, and Value matrices multiplication
- Scaled Dot-Product Attention
- Softmax normalization to compute attention scores

### CODE AND OUTPUT

```
import numpy as np
import nltk
from nltk.tokenize import word_tokenize

# Ensure NLTK tokenizer is ready
nltk.download('punkt')

# Step 1: Tokenize the sentence
sentence = "The quick brown fox jumped over the lazy dog."
tokens = word_tokenize(sentence)
n = len(tokens) # Number of tokens
d_k = 8 # Embedding size (for simplicity, we use 8)

# Step 2: Create random embeddings for each token
np.random.seed(42)
embeddings = np.random.rand(n, d_k)

# Step 3: Initialize random Q, K, V matrices
W_q = np.random.rand(d_k, d_k)
W_k = np.random.rand(d_k, d_k)
W_v = np.random.rand(d_k, d_k)

Q = embeddings @ W_q
K = embeddings @ W_k
V = embeddings @ W_v

# Step 4: Compute Attention Scores
scores = Q @ K.T / np.sqrt(d_k) # Scaled dot product
attention_weights = np.exp(scores) / np.sum(np.exp(scores), axis=1, keepdims=True) #
Softmax
```

```
# Step 5: Compute final attention output
attention_output = attention_weights @ V

# Display the attention scores
import pandas as pd
attention_df = pd.DataFrame(attention_weights, index=tokens, columns=tokens)
print("\nSelf-Attention Scores:")
print(attention_df)
```

Self-Attention Scores:

|        | The      | quick    | brown    | fox      | jumped   | over     | the \    |
|--------|----------|----------|----------|----------|----------|----------|----------|
| The    | 0.003225 | 0.002700 | 0.000757 | 0.002709 | 0.024326 | 0.001251 | 0.953374 |
| quick  | 0.002369 | 0.002424 | 0.000576 | 0.002138 | 0.018002 | 0.000943 | 0.964699 |
| brown  | 0.008734 | 0.008929 | 0.002862 | 0.008271 | 0.045409 | 0.004289 | 0.890832 |
| fox    | 0.005147 | 0.005647 | 0.001526 | 0.005055 | 0.032319 | 0.002317 | 0.928991 |
| jumped | 0.002577 | 0.002410 | 0.000602 | 0.002337 | 0.021735 | 0.000944 | 0.959645 |
| over   | 0.005734 | 0.005728 | 0.001688 | 0.005175 | 0.032849 | 0.002603 | 0.926154 |
| the    | 0.000150 | 0.000173 | 0.000019 | 0.000144 | 0.003588 | 0.000041 | 0.994994 |
| lazy   | 0.018685 | 0.021674 | 0.007699 | 0.018702 | 0.071094 | 0.010724 | 0.788503 |
| dog    | 0.003494 | 0.004398 | 0.001055 | 0.003891 | 0.027209 | 0.001569 | 0.943473 |
| .      | 0.004524 | 0.004261 | 0.001231 | 0.004050 | 0.029131 | 0.001863 | 0.939130 |

|        | lazy     | dog      | .        |
|--------|----------|----------|----------|
| The    | 0.000534 | 0.007574 | 0.003550 |
| quick  | 0.000377 | 0.005678 | 0.002793 |
| brown  | 0.002073 | 0.018180 | 0.010420 |
| fox    | 0.001044 | 0.011533 | 0.006421 |
| jumped | 0.000395 | 0.006246 | 0.003108 |
| over   | 0.001167 | 0.012250 | 0.006652 |
| the    | 0.000010 | 0.000642 | 0.000237 |
| lazy   | 0.005555 | 0.034363 | 0.023000 |
| dog    | 0.000696 | 0.009016 | 0.005198 |
| .      | 0.000843 | 0.009794 | 0.005173 |

## INFERENCE

The self-attention mechanism assigns varying attention scores to words in the sentence.

Words with stronger contextual relevance receive higher attention scores.

This demonstrates how self-attention can capture dependencies across words in a sequence.